27-03-2026

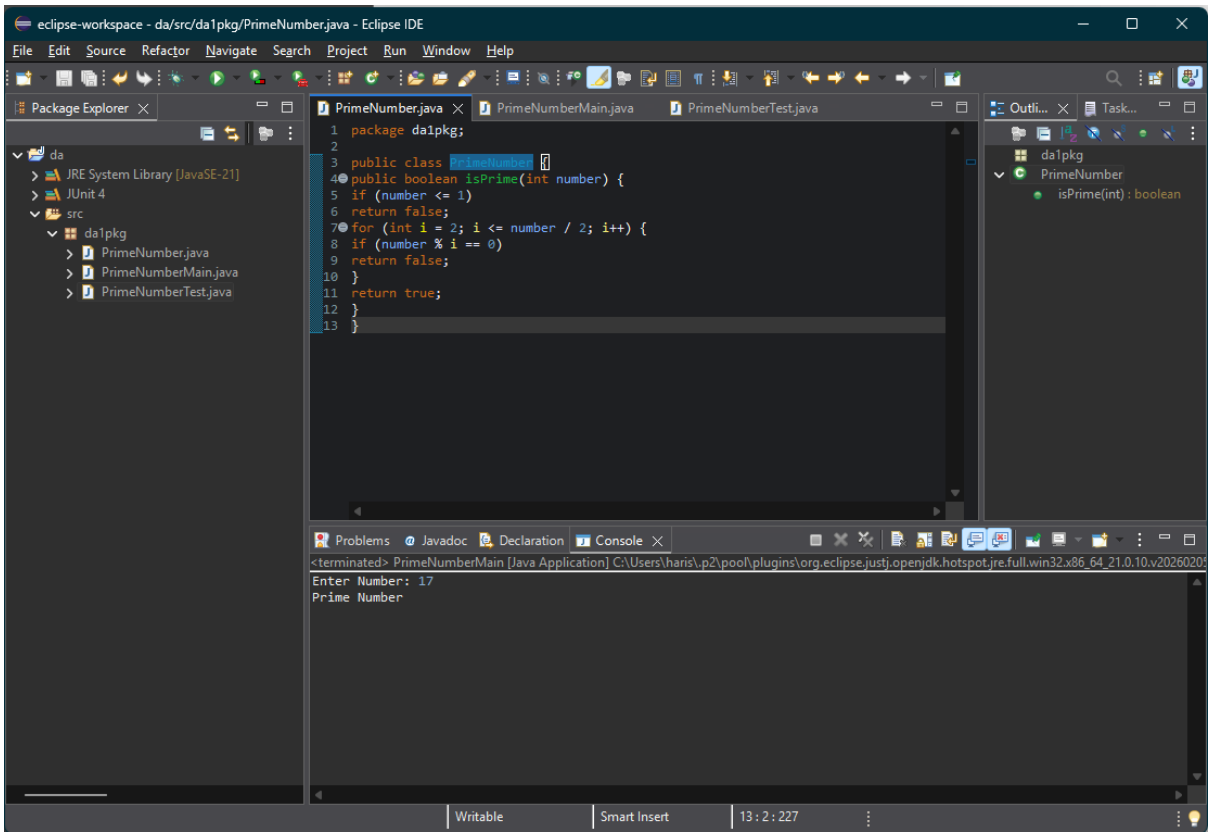
Problem 1: PRIME NUMBER CHECK

Problem Statement: Write a Java program to check whether a number is Prime or Not.
Conditions:

- Number should be greater than 1
- If divisible by any number other than 1 and itself → Not Prime

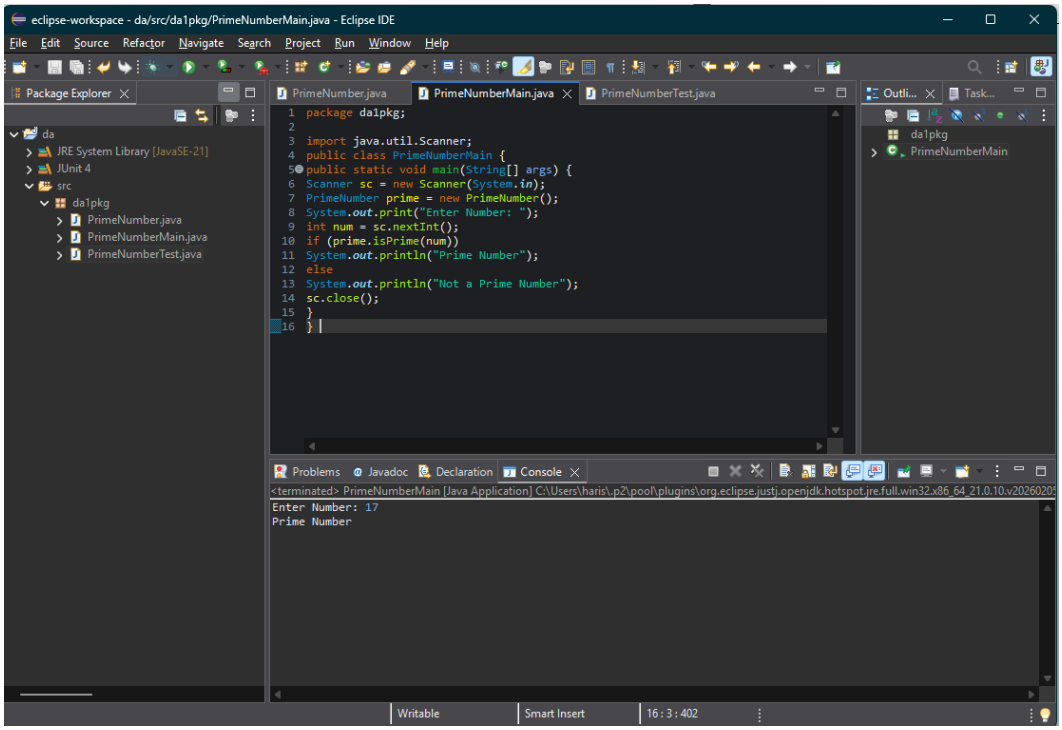
PRIME NUMBER CHECK

Create New Class (PrimeNumber)

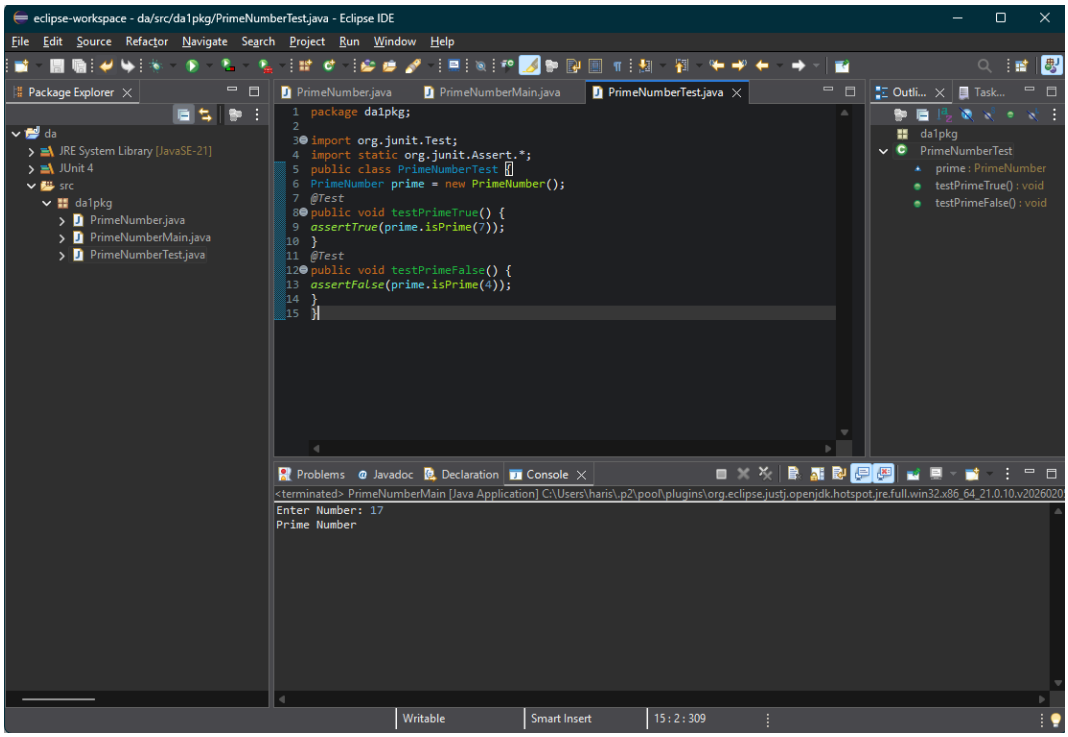


27-03-2026

Create New Class (PrimeNumberMain)



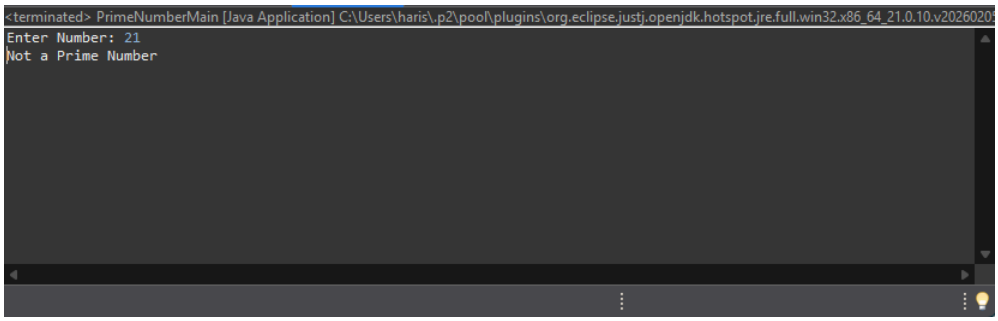
Create New Junit Test Case (PrimeNumberTest)

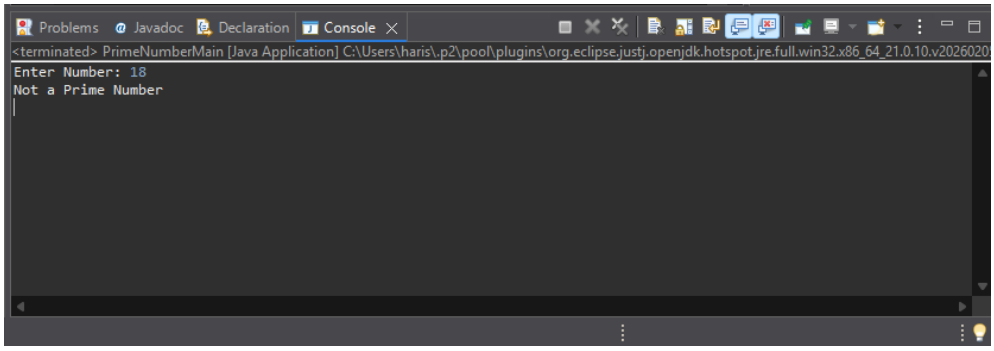


27-03-2026

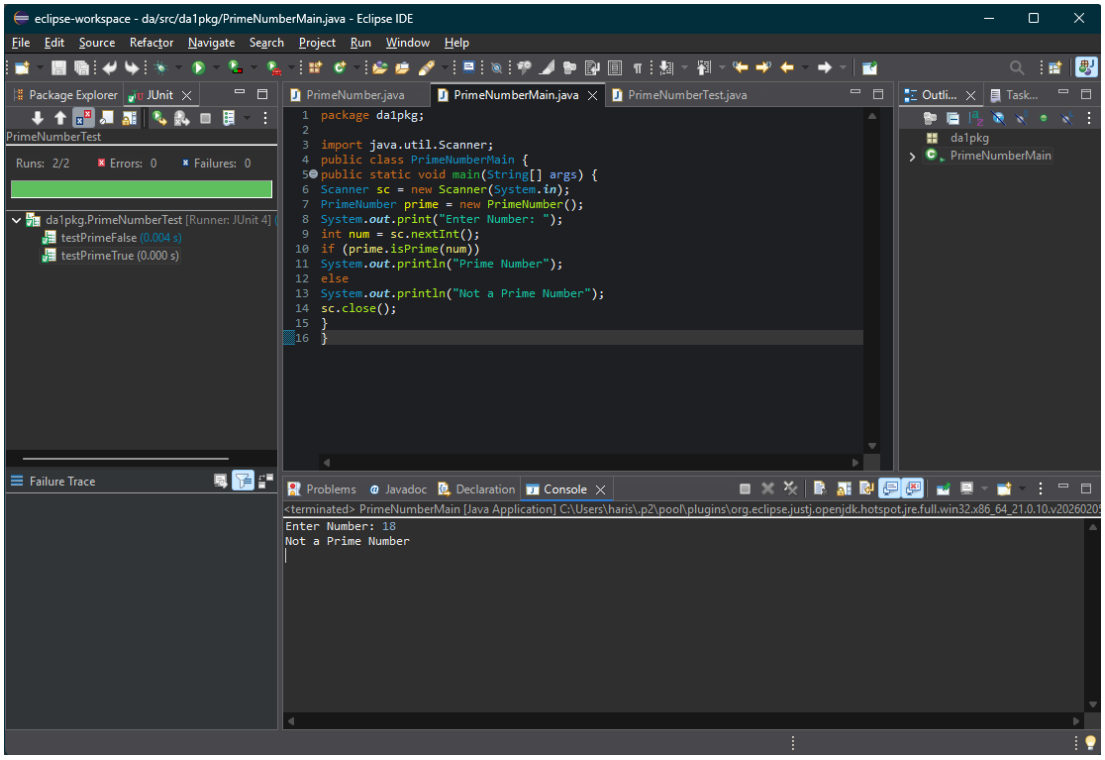
RUNNING TESTCASE

Main Class Output





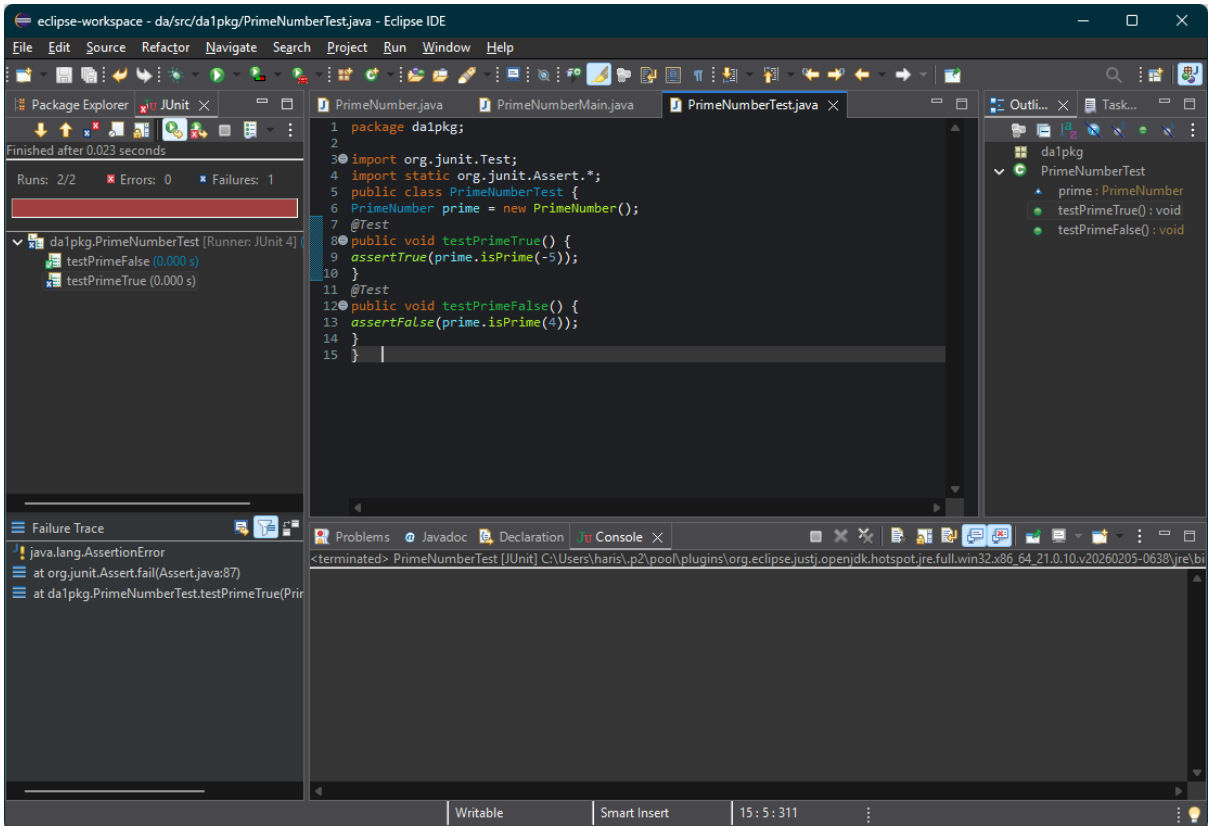
Junit Output



27-03-2026

Negative Testcase

assertTrue(prime.isPrime(-5)); //which is not prime number here and provides negative false testcase



Problem 2. FACTORIAL CALCULATION

Problem Statement: Write a Java program to calculate factorial of a number.
Conditions:

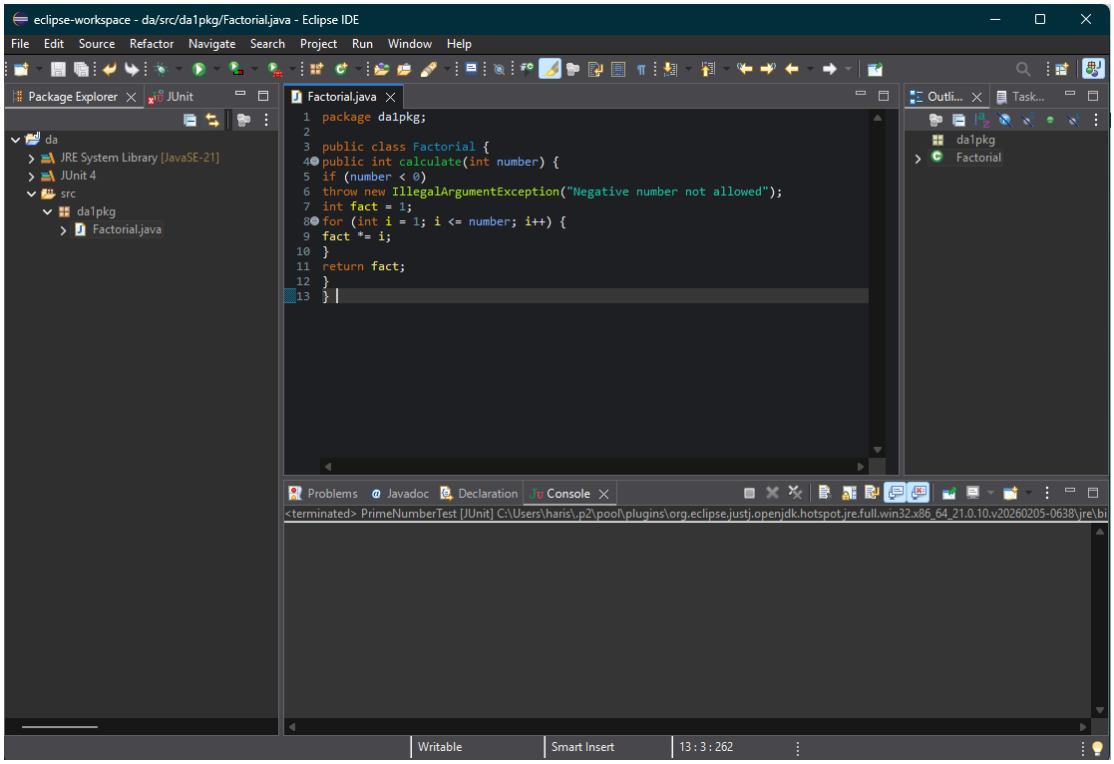
- Factorial of negative number → throw Illegal Argument Exception

- 0! = 1

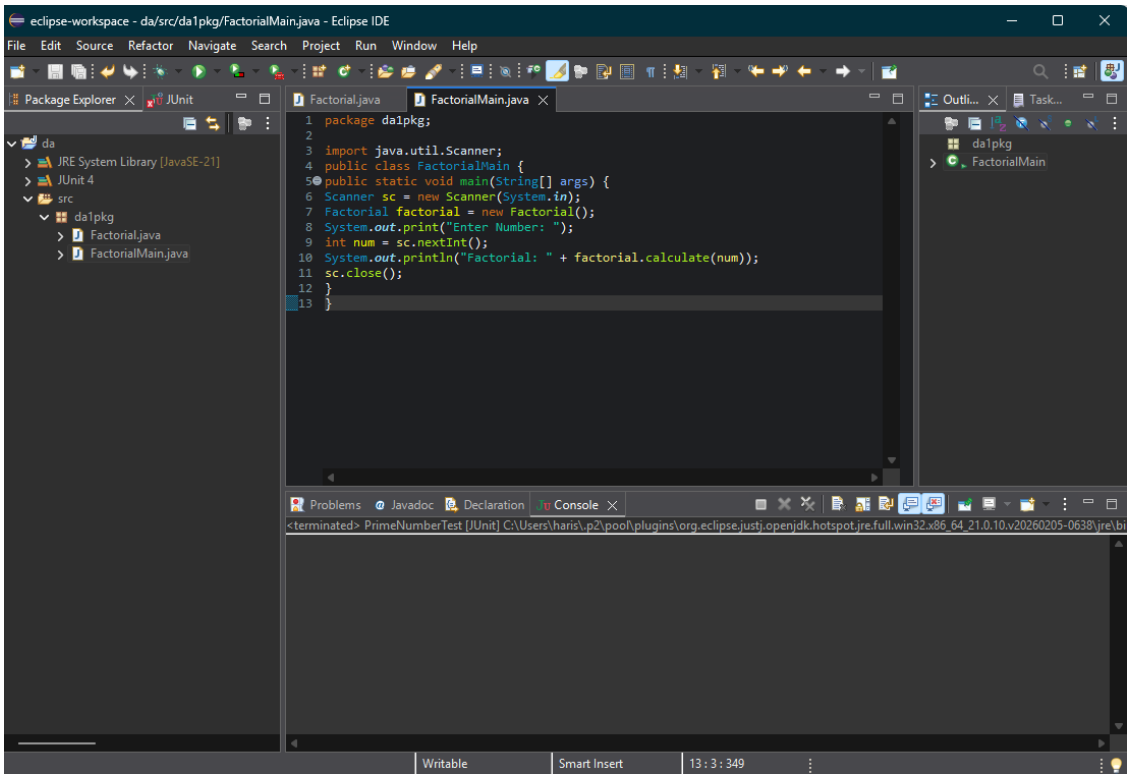
27-03-2026

FACTORIAL CALCULATION

Create New Class (Factorial)

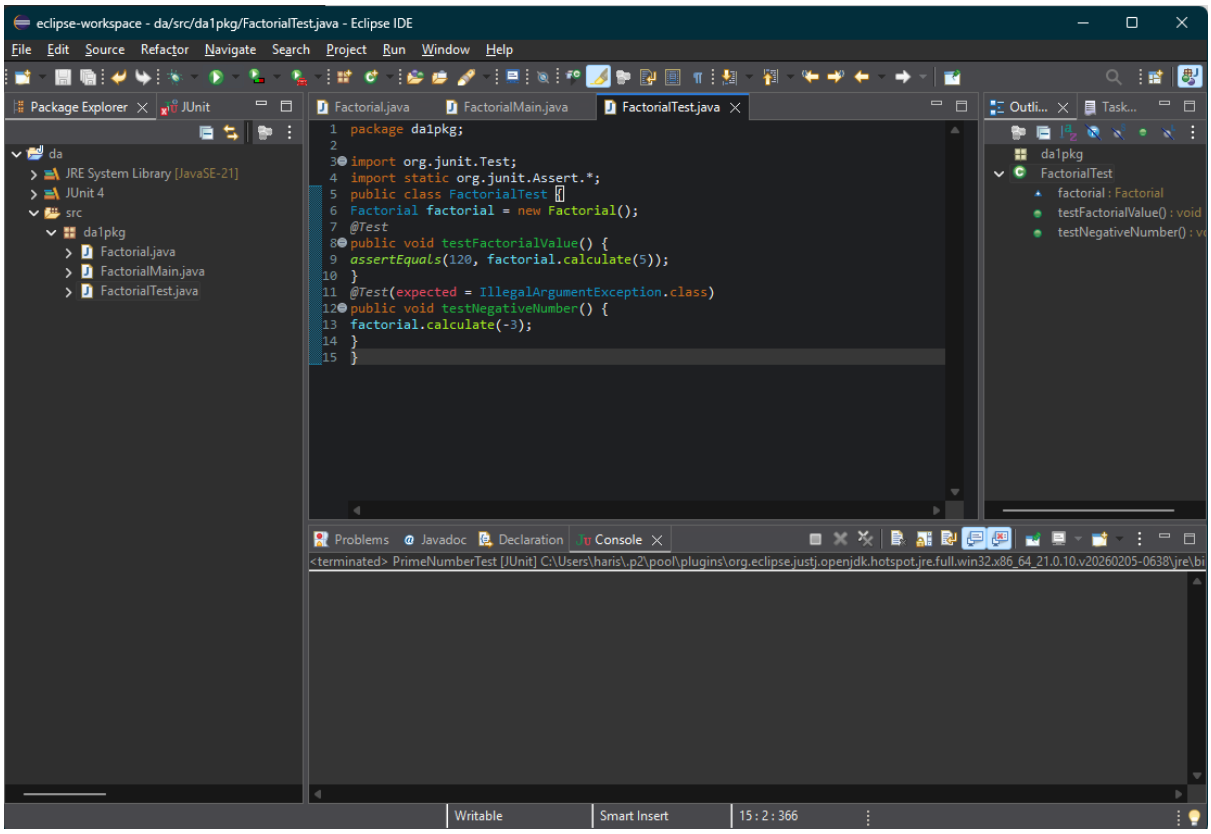


Create New Class (FactorialMain)



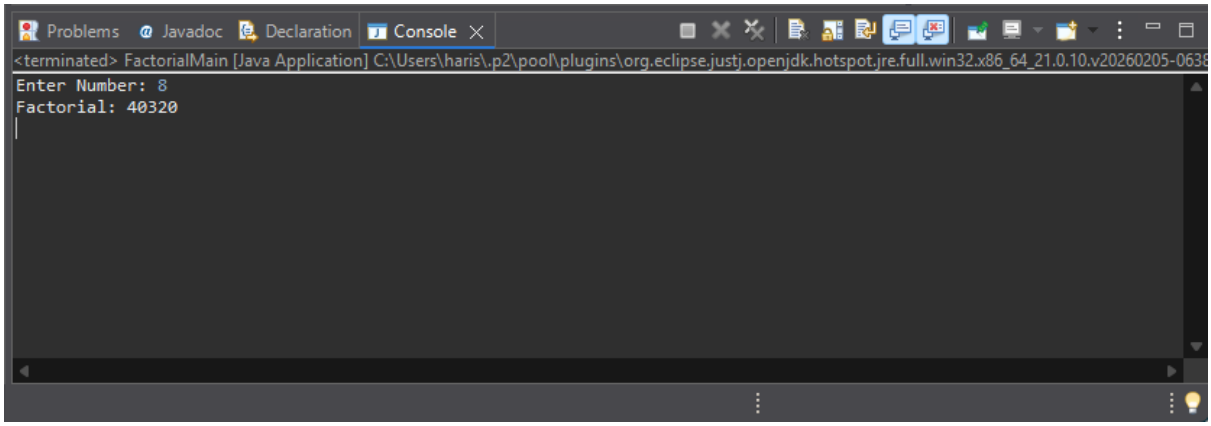
27-03-2026

Create New Junit Test Case (PrimeNumberTest)

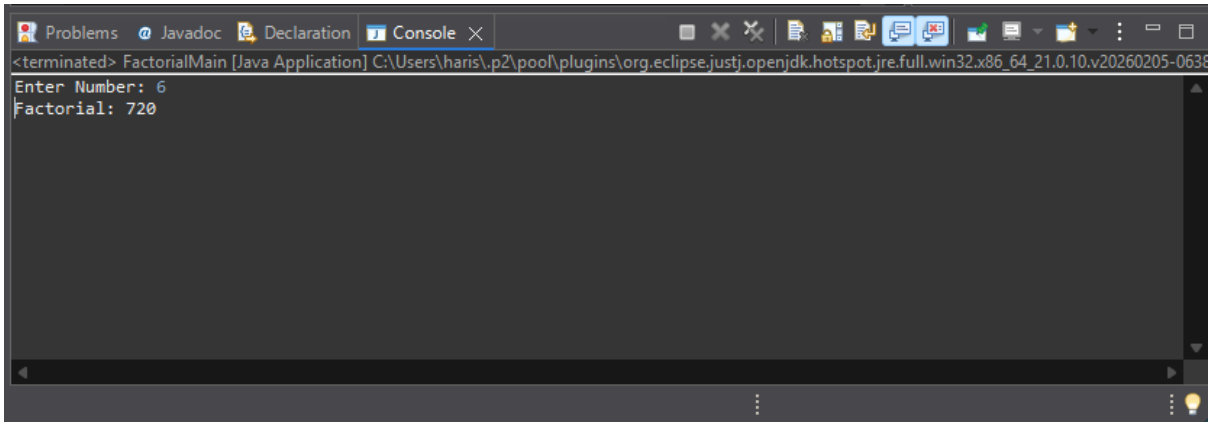


RUNNING TESTCASE

Main Class Output

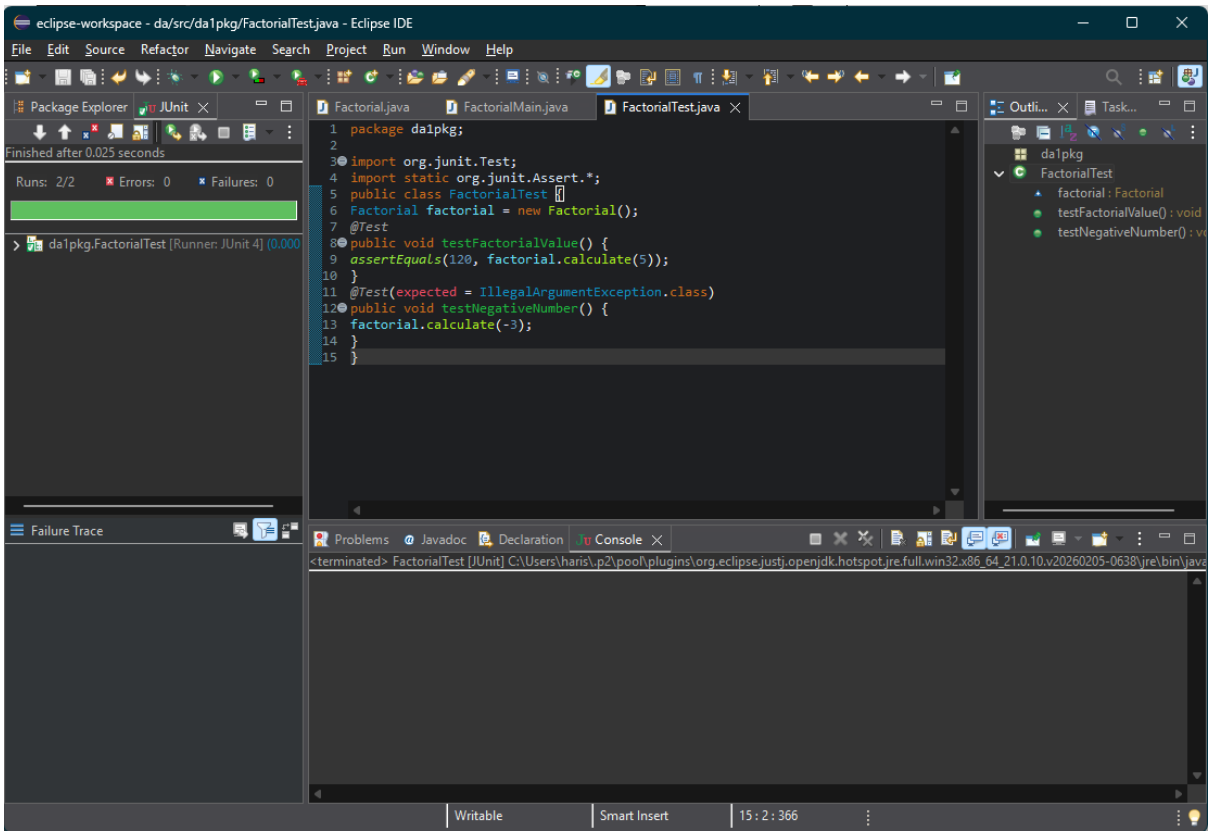


27-03-2026

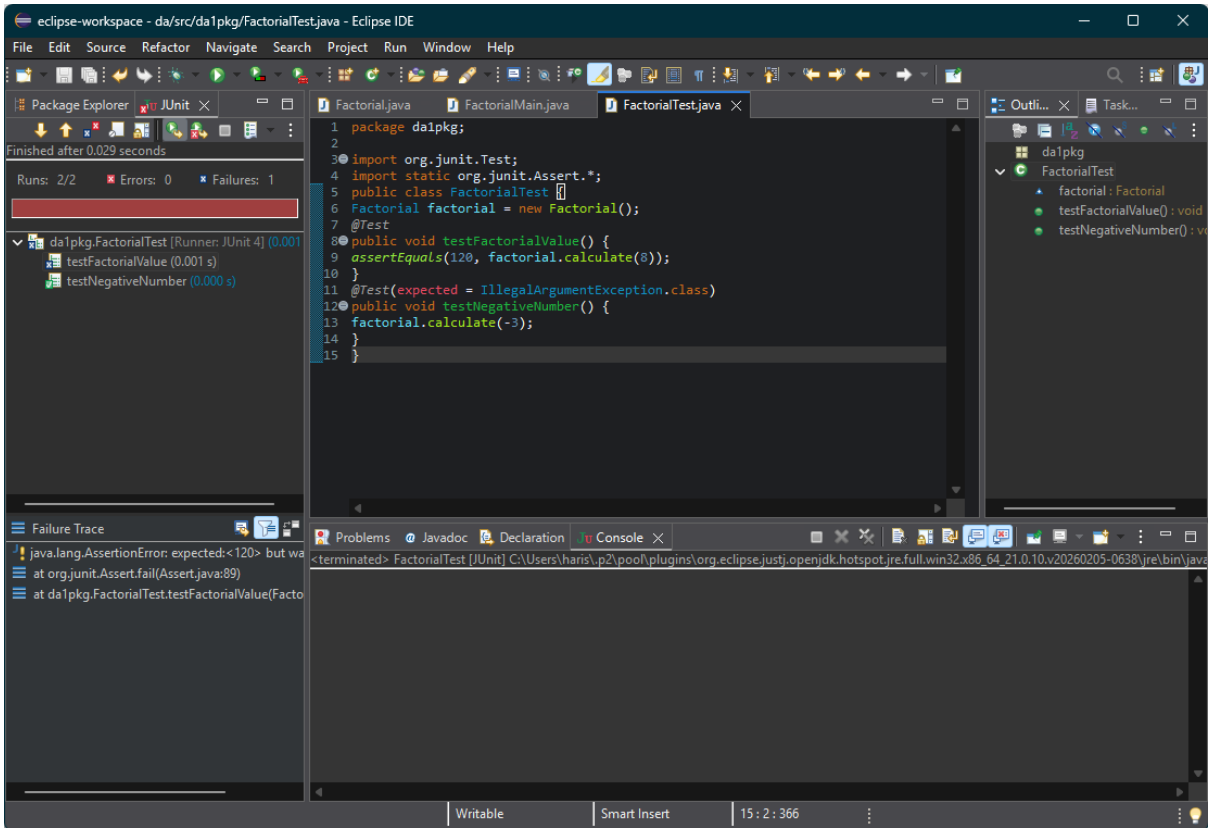


Junit Output

assertEquals(120, factorial.calculate(5)); //true condition



27-03-2026

Negative Testcase

assertEquals(120, factorial.calculate(8)); //false condition